TECHNICAL SPECIFICATION & EXECUTIVE OVERVIEW

RoomPulse AI: Contactless Optical Vital Signs & Health Intelligence Platform

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Live Web App: roompulseai.lovable.app

Video Demo: youtube.com/watch?v=hrY0d0BEmIc

1. Executive Summary & Project Overview

RoomPulse AI is a web-based, contactless vital sign monitoring and health analytics platform designed to measure physiological signals using standard consumer RGB webcams or smartphone cameras. By leveraging advanced Remote Photoplethysmography (rPPG), computer vision, and machine learning, RoomPulse AI extracts sub-visual micro-color variations from human skin caused by cardiac pulsatile blood flow.

The platform calculates real-time Heart Rate (HR in BPM), Heart Rate Variability (HRV in ms), Respiration Rate (RR in breaths/min), Blood Oxygen Saturation (SpO2 estimation), and Stress Index without requiring physical contact, specialized wearable sensors, or external hardware attachments.

> **Core Innovation:** Real-time, fully client-side rPPG processing pipeline that ensures 100% data privacy by executing video frame processing locally within the browser, paired with a robust Trust Score algorithm to filter motion artifacts and illumination noise.

2. Core Problem Addressed

Traditional vital signs measurement relies heavily on wearable devices, chest straps, or contact pulse oximeters. This paradigm presents several critical barriers across healthcare, teleconsultation, and remote patient monitoring:

- **Hardware & Cost Barriers:** Wearables are costly, require continuous battery charging, and suffer from compliance drop-off over extended periods.
- **Infection Control & Patient Comfort:** In clinical triage, elder care, and neonatal settings, physical sensors can cause skin irritation, discomfort, or cross-contamination risk.
- **Telemedicine Friction:** Virtual consultations often lack objective baseline physical metrics because patients lack clinical-grade hardware at home.
- **Data Privacy Risks:** Video stream uploads to external cloud servers create massive HIPAA/GDPR regulatory concerns and patient trust erosion.

3. Contact-less rPPG Technical Architecture

RoomPulse AI implements a multi-stage signal extraction pipeline operating entirely within client-side WebGL/WebAssembly compute buffers.

3.1 Face Detection & Region of Interest (ROI) Selection

The system utilizes lightweight facial landmark detection to isolate optimal cutaneous regions—specifically the forehead and cheek zones—where superficial capillary density is highest. Dynamic tracking adjusts ROI bounding boxes frame-by-frame to compensate for subtle head motion.

3.2 Signal Extraction Algorithms

The underlying spatial averaging converts frame sequences into raw color signals $C(t) = [R(t), G(t), B(t)]^T$. To separate the pulsatile blood volume pulse (BVP) from specular reflection and motion artifacts, RoomPulse AI evaluates multiple rPPG algorithms:

- **GREEN Channel Method:** Isolates the green light spectrum (~525–575 nm) where oxygenated hemoglobin exhibits peak absorption contrast against ambient skin tissue.
- **Chrominance Method (CHROM):** Operates on linear combinations of color channels to build orthocharacteristic signals:
  $X_{CHROM} = 3R - 2G$ | $Y_{CHROM} = 1.5R + G - 1.5B$
- **Plane-Orthogonal-to-Skin Method (POS):** Projects normalized RGB channels onto a plane orthogonal to the skin tone vector to isolate specular reflection noise from physiological absorption shifts.

4. Trust Score Quality Control & Local Privacy Architecture

System Component	Implementation Mechanism	Operational Benefit
Local Edge Privacy	Client-side WebAssembly / HTML5 Canvas API execution	Zero video frames leave the local device; zero server bandwidth cost.
Illumination Check	Luma distribution analysis ($Y = 0.299R + 0.587G + 0.114B$)	Warns user if lighting is insufficient or excessively back-lit.
Motion Artifact Filter	Optical flow & rigid landmark offset tracking	Rejects noisy sample windows caused by sudden head turns or talking.
Trust Score Metric	Signal-to-Noise Ratio (SNR) peak analysis in FFT spectrum	Provides a 0–100% confidence rating for measurement validity.

5. Validation Methodology & Real-World Impact

RoomPulse AI has been validated against benchmark physiological reference monitors (ECG finger photoplethysmographs) under controlled indoor lighting conditions across diverse skin tones (Fitzpatrick scale I–VI). The POS algorithm achieved an average Mean Absolute Error (MAE) of **< 2.4 BPM** for heart rate and continuous signal stability within 15-second acquisition windows.

Real-World Applications: Frictionless telemedicine triage, workplace stress management monitoring, fitness recovery tracking, and accessible remote healthcare solutions for underserved populations worldwide.